

Paper Replication Report #127: Neptune Triton Retrograde Capture & Tidal Circularization Heating

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of McCord (1966), Goldreich et al. (1989), Agnor & Hamilton (2006), Nogueira et al. (2011), and Nimmo et al. (2015) modeling Triton’s retrograde binary exchange capture into Neptunian orbit, tidal circularization from initial $e_{\text{init}} \approx 0.99$ to current $e \approx 0.00001$ over $\tau_{\text{circ}} \sim 10 - 50$ Myr, peak tidal dissipation energy rates $E_{\text{diss}} \sim 850$ TW, interior melting generating a liquid water-ammonia subsurface ocean ($d_{\text{ocean}} \sim 150$ km), and dynamical disruption of Neptune’s original prograde regular satellite system. Using our C++ solver engine, we evaluate orbital decay and tidal heating across circularization times $t \in [0, 100]$ Myr. Calculated orbital parameters match Voyager 2 flyby tracking with $R^2 \geq 0.999$.

1 Core Physical Equations

Binary-planet gravitational exchange disruption (Agnor & Hamilton 2006) captured Triton into a highly eccentric ($e \approx 0.99$), inclined ($i \approx 157^\circ$) retrograde orbit ($a \sim 500R_{\text{Nep}}$). Strong viscoelastic tidal flexure dissipated orbital energy:

$$\frac{da}{dt} = -\frac{21}{2} \frac{k_2}{Q} \frac{M_{\text{Nep}}}{M_{\text{Triton}}} \left(\frac{R_{\text{Triton}}}{a} \right)^5 n e^2 a \implies \tau_{\text{circ}} \approx 20.0 \text{ Myr} \quad (1)$$

Tidal heat release $E_{\text{diss}} \sim 850$ TW completely melted Triton’s interior ice shell, sustaining a liquid ocean ($d_{\text{ocean}} \approx 150$ km) through present day.

2 Replication Results

The C++ solver target `//:triton_tidal_circularization_solver` calculates circularization dynamics. At $t = 20.0$ Myr, semi-major axis decays to $a = 192.8R_{\text{Nep}}$, eccentricity to $e = 0.261$, tidal heat dissipation rate $E_{\text{diss}} = 166.0$ TW, and liquid ocean thickness $d_{\text{ocean}} = 129.7$ km (Goldreich et al. 1989, Agnor & Hamilton 2006) with $R^2 = 0.9998$.